

ML-Ops Final Project Design Doc

Dynamic Discounting model evaluation app: A Command Line Interface App For evaluating sales forecasting models

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Purpose

The Dynamic Discounting (DD) project is a price optimization initiative for a fashion retail brand in India. At its core, it is a data-driven project involving multiple stages — including effective product categorization, category-level sales forecasting, forecasting model evaluation, scenario analysis, and ultimately an optimization phase that allocates ideal prices to products.

While each of these steps is important, the forecasting stage plays a pivotal role in determining the success of the overall dynamic discounting strategy. In today's machine learning ecosystem, a wide range of techniques can be applied to forecasting tasks — from basic linear regression and SARIMA models to advanced boosting methods and deep learning architectures. Given the diversity of model types and the varying KPIs across teams and organizations, the process of model selection can often become unstructured and inconsistent.

To address this, we introduce the Dynamic Discounting Model Evaluation App (DDMEA) — a command-line interface (CLI) tool designed to streamline model evaluation and comparison. The app integrates into the current workflow of ML engineers, where models are saved as pickle files and evaluated on a validation set representing a previously unseen future time span.

The tool computes and displays key evaluation metrics, including:

- Total Products Evaluated
- Total Predictions
- Mean Absolute Error (MAE)
- Median Absolute Error
- Mean Percent Error
- Median Percent Error
- Total Actual Sales
- Total Predicted Sales
- Overall Forecasting Error

By providing a simple yet effective interface for model evaluation, DDMEA enables analysts and ML engineers to store, run, evaluate, and compare forecasting models more systematically — helping them make informed choices about model performance in the context of dynamic pricing.

Ideal User

DDMEA is an internal tool specifically designed for use within the DD project. The ideal users are analysts and machine learning engineers who are actively involved in the DD initiative and are familiar with its workflows and objectives.

These users are expected to have:

- Proficiency with basic command-line operations such as directory navigation.
- A working understanding of forecasting models and the statistical evaluation metrics used in the app (e.g., MAE, percent error, total predicted sales).
- Experience building and saving models (e.g., as .pkl files) and running them on validation datasets.

Important note: If a user's model involves deep learning techniques that require GPU acceleration, they must ensure their machine has a compatible GPU setup to run the model evaluations effectively.

Tasks

To effectively use DDMEA, the user must complete a few prerequisites and will be able to perform key tasks within the tool. These include:

- Prerequisites
 - The user must have a pre-trained forecasting model saved as a .pkl file.
 - The model's corresponding evaluation function should be implemented and stored in a .py file.
 - Both files must be placed in the appropriate directories, as specified in the project's README.
- Core Tasks
 - **Run Model Evaluations:** Users can run their saved models on the validation dataset. If the model's outputs have not been previously generated, DDMEA will compute and store them.
 - **Store Evaluation Results:** Model performance metrics and visualisations are saved systematically for future reference and comparison.
 - **Compare Models:** Users can select and compare any two models on a range of evaluation metrics to determine which performs better under the criteria relevant to the DD projects

Workflow

The DDMEA tool provides a simple command-line interface that guides users through the process of running and evaluating forecasting models — all from a single terminal session. The workflow is designed to be intuitive and structured, enabling analysts and ML engineers to focus on model performance rather than setup.

Here's how it works:

1. Startup & Setup Checks

Upon launching the app, DDMEA scans the relevant directories (models/ and evaluation_scripts/) to detect available model files (.pkl) and their corresponding evaluation scripts (.py). It dynamically builds a menu of runnable options, ensuring that only properly configured models appear.

2. Prerequisite Reminder

The tool reminds users of the basic folder structure and prerequisites:

- Ensure the model is saved in the correct format and directory.
- Ensure the evaluation function script is correctly linked and functional.

3. Model Execution

Users are presented with a menu of models to run. Selecting a model executes its evaluation script and generates the performance metrics (if not already done). These results are stored for future access.

4. Validation Checks

Before allowing comparison, DDMEA checks whether valid output (forecast_evaluation_details.csv) exists for each model. If any model lacks a valid result, the user is prompted to run it first.

5. Model Comparison

Once all models have been evaluated successfully, users are given the option to run a comparison script (compare_evals.py) to directly compare performance across models using key metrics like MAE, total predicted vs actual sales, and percent errors.

6. Repeat or Exit

After running a model or comparison, users can choose to evaluate another model or exit the app.